



# AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH (AIUB)

Faculty of Science and Technology (FST)  
Department of Computer Science (CS)  
Undergraduate Program

| COURSE PLAN                                                                                                                                                                                                                                                                                                                        | SEMESTER: Summer 2025-2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>I. Course Core and Title</b><br/>CSC 4233 Natural Language Processing</p> <p><b>II. Credit</b><br/>3 credit hours (3 hours of theory per week)</p> <p><b>III. Nature</b><br/>Elective Course for CS, CSE, CSSE, SE, CIS</p> <p><b>IV. Prerequisite</b><br/>CSC 3216 Compiler Design &amp; CSC 4262 Programming in Python</p> | <p><b>V. Vision:</b><br/>Our vision is to be the preeminent Department of Computer Science through creating recognized professionals who will provide innovative solutions by leveraging contemporary research methods and development techniques of computing that is in line with the national and global context.</p> <p><b>VI. Mission:</b><br/>The mission of the Department of Computer Science of AIUB is to educate students in a student-centric dynamic learning environment; to provide advanced facilities for conducting innovative research and development to meet the challenges of the modern era of computing, and to motivate them towards a life-long learning process.</p> |

## VII - Course Description

Natural Language Processing (NLP) is a core area of Artificial Intelligence concerned with enabling computers to understand, process, and generate human language. This course provides a comprehensive, practice-oriented introduction to NLP, progressing systematically from classical statistical methods to modern deep learning. Students will gain hands-on experience with Python-based NLP toolkits (NLTK, spaCy, HuggingFace Transformers), work with both English and Bangla text, and complete a substantial group project deploying an NLP system. Upon completion of this course, students will be able to:

- Understanding the basics of natural language processing.
- Apply text preprocessing, tokenisation, and normalisation pipelines to raw text data.
- Explore text classification techniques and their evaluation metrics.
- Build and evaluate statistical and neural language models for NLP tasks such as sentiment analysis.
- Understanding text representations for applying natural language processing approaches.
- Explore text similarity method, language models & neural language models.
- Use pretrained models (BERT, GPT) with fine-tuning strategies for downstream classification, NER, and QA.
- Design and implement an end-to-end NLP project with a working demo to illustrate how the learned topics can be applied to real-life applications .

## VIII – Course outcomes (CO) Matrix

By the end of this course, students should be able to:

| COs*  | CO Description                                                                                                                                                                             | Level of Domain*** |   |   | PO Assessed**** |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|---|---|-----------------|
|       |                                                                                                                                                                                            | C                  | P | A |                 |
| CO1** | <b>Apply</b> the fundamental concepts of NLP — including text preprocessing pipeline, statistical language models, word representations (BoW, TF-IDF, embeddings), and evaluation metrics. | 3                  |   |   | PO-a4.C3        |
| CO2   | <b>Use</b> suitable pre-trained models to develop appropriate modern language models that solve complex NLP problems.                                                                      |                    | 4 |   | PO-e2.P4        |

*C: Cognitive; P: Psychomotor; A: Affective Domain*

\* *CO assessment method and rubric of COs assessment is provided in later section*

\*\* *COs will be mapped with the Program Outcomes (POs) for PO attainment*

\*\*\* *The numbers under the 'Level of Domain' columns represent the level of Bloom's Taxonomy each CO corresponds to.*

\*\*\*\* *The numbers under 'PO Assessed' column represent the POs each CO corresponds to.*

## IX – Topics to be covered in the class and/or lab\*

| Time Frame                                | CO Mapped | Topics                                                                                                 | Teaching Activities      | Assessment Strategy(s)   |
|-------------------------------------------|-----------|--------------------------------------------------------------------------------------------------------|--------------------------|--------------------------|
| Week 1                                    | CO1       | OBE Discussion. Introduction to NLP, Regular expressions                                               | Lecture, Question-answer | Assignment/Project, Quiz |
| Week 2                                    | CO1       | Tokenization, Case folding, Synonym substitution, Stemming, Lemmatization                              | Lecture, Question-answer | Assignment/Project, Quiz |
| Week 3                                    | CO1       | Punctuation removal, Stop words removal, Text Classification: Basics, Sentiment analysis, Bag of Words | Lecture, Question-answer | Assignment/Project, Quiz |
| Week 4                                    | CO1       | Text Classification: Naïve Bayes classifier, Evaluation                                                | Lecture, Question-answer | Assignment/Project, Quiz |
| Week 5                                    | CO1       | Vector semantics and Embeddings                                                                        | Lecture, Question-answer | Assignment/Project, Quiz |
| Week 6                                    | CO1       | Midterm review, Project discussion, Brainstorming                                                      | Lecture, Question-answer | Assignment/Project       |
| Midterm (Week 7)                          |           |                                                                                                        |                          |                          |
| Week 8                                    | CO2       | Edit distance, N-Gram language models                                                                  | Lecture, Question-answer | Assignment/Project, Quiz |
| Week 9                                    | CO2       | N-Gram language models (continue), Neural networks & Neural language models                            | Lecture, Question-answer | Assignment/Project, Quiz |
| Week 10                                   | CO2       | Neural networks & Neural language models (continue)                                                    | Lecture, Question-answer | Assignment/Project, Quiz |
| Week 11                                   | CO2       | Transformers                                                                                           | Lecture, Question-answer | Assignment/Project, Quiz |
| Week 12                                   | CO2       | Transformers (continue)                                                                                | Lecture, Question-answer | Assignment/Project, Quiz |
| Week 13                                   | CO2       | Final term review, Project discussion, Brainstorming.                                                  | Lecture, Question-answer | Assignment/Project       |
| Final term (Week 14)                      |           |                                                                                                        |                          |                          |
| Project Submission/Defense/Viva (Week 15) |           |                                                                                                        |                          |                          |

\* *The faculty reserves the right to change, amend, add or delete any of the contents.*

## X – Mapping of PO/PLO and K, P, A of this course:

| PO Indicator ID | PO Indicators Definition (As per the requirement of WKs)                                                                        | Domain                       | K  | P              | A |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------|------------------------------|----|----------------|---|
| PO-a4.C3        | Apply information and concepts in specialized engineering sciences with the in-depth analysis of a complex engineering problem. | Cognitive Level 3 (Applying) | K4 | P1<br>P3<br>P7 |   |

|          |                                                                                                                                                     |                                          |  |                |  |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|--|----------------|--|
| PO-e2.P4 | Use modern engineering and IT tools for prediction and modeling of complex engineering problems considering the practice in engineering discipline. | Psychomotor<br>Level 4<br>(Articulation) |  | P1<br>P4<br>P5 |  |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|--|----------------|--|

## XI – K, P, A Definitions

| Indicator | Title                                       | Description                                                                                                                                                                                             |
|-----------|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| K4        | Forefront specialist knowledge for practice | Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline |
| P1        | Depth of knowledge required                 | Cannot be resolved without in-depth engineering knowledge at the level of one or more of K3, K4, K5, K6 or K8 which allows a fundamentals-based, first principles analytical approach.                  |
| P3        | Depth of analysis required                  | Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models.                                                                                           |
| P4        | Familiarity of issues                       | Involve infrequently encountered issues                                                                                                                                                                 |
| P5        | Extent of applicable codes                  | Are outside problems encompassed by standards and codes of practice for professional engineering                                                                                                        |
| P7        | Interdependence                             | Are high level problems including many component parts or sub-problems.                                                                                                                                 |

## XII – Mapping of CO Assessment Method and Rubric

The mapping between Course Outcome(s) (COs) and the Selected Assessment method(s) and the mapping between Assessment method(s) and Evaluation Rubric(s) is shown below:

| COs | Description                                                                                                                                                                                | Mapped POs | Assessment Method    | Assessment Rubric               |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------------------|---------------------------------|
| CO1 | <b>Apply</b> the fundamental concepts of NLP — including text preprocessing pipeline, statistical language models, word representations (BoW, TF-IDF, embeddings), and evaluation metrics. | PO-a4.C3   | Assignment / Project | Rubric for Assignment / Project |
| CO2 | <b>Use</b> suitable pre-trained models to develop appropriate modern language models that solve complex NLP problems.                                                                      | PO-e2.P4   | Assignment / Project | Rubric for Assignment / Project |

## XIII – Evaluation and Assessment Criteria

| CO1: Apply the fundamental concepts of NLP — including text preprocessing pipeline, statistical language models, word representations (BoW, TF-IDF, embeddings), and evaluation metrics. |                                                           |                                                                                         |                                                      |                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------------------------------------|------------------------------------------------------|---------------------------------------------------|
| Assessment Attribute/Criteria                                                                                                                                                            | Missing/ Incorrect (0)                                    | Inadequate (1-4)                                                                        | Satisfactory (5-8)                                   | Excellent (9-10)                                  |
| <b>Applied required concepts appropriately</b>                                                                                                                                           | Not applied or applied the required concepts incorrectly. | The concepts are understood, but applied them to the given problem with major mistakes. | The required concepts are applied with a few errors. | All the concepts are applied in appropriate ways. |

**CO2: Use suitable pre-trained models to develop appropriate modern language models that solve complex NLP problems.**

| Assessment Criteria            | Missing/ Incorrect (0)       | Inadequate (1-4)                                                                     | Satisfactory (5-8)                                        | Excellent (9-10)                                               |
|--------------------------------|------------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------------------|----------------------------------------------------------------|
| <b>Accurate implementation</b> | No/incorrect implementation. | The implementation is not sufficient for the complete solution of the given problem. | A correct implementation is submitted with little issues. | The given implementation fulfil all the requirements properly. |

**XIV- Course Requirements**

- Attending at least 80% of the class.
- Submission of projects and related materials in due time.

**XV – Evaluation & Grading System\***

The following grading system will be strictly followed in this class:

| MID TERM                                                     |             | FINAL TERM               |             |
|--------------------------------------------------------------|-------------|--------------------------|-------------|
| Attendance                                                   | 10%         | Attendance               | 10%         |
| Quiz                                                         | 40%         | Quiz                     | 40%         |
| Project, Report and Viva                                     | 50%         | Project, Report and Viva | 50%         |
| <b>Total</b>                                                 | <b>100%</b> | <b>Total</b>             | <b>100%</b> |
| <b>Grand Total 100% = 40% of Midterm + 60% of Final Term</b> |             |                          |             |

| Letter | Grade Point | Numerical %             |
|--------|-------------|-------------------------|
| A+     | 4.00        | 90-100                  |
| A      | 3.75        | 85 - < 90               |
| B+     | 3.50        | 80 - < 85               |
| B      | 3.25        | 75 - < 80               |
| C+     | 3.00        | 70 - < 75               |
| C      | 2.75        | 65 - < 70               |
| D+     | 2.50        | 60 - < 65               |
| D      | 2.25        | 50 - < 60               |
| F      | 0.00        | < 50                    |
| I      |             | Incomplete              |
| W      |             | Withdrawal              |
| UW     |             | Unofficially Withdrawal |

\* The evaluation system will be strictly followed as per the AIUB grading policy.

\* CO attainment will be achieved with 60% of the evaluation marks.

**XII – Textbook/ References**

1. Jurafsky & Martin — Speech and Language Processing (3rd ed.)
2. Goldberg — A Primer on Neural Network Models for NLP
3. Tunstall et al. — Natural Language Processing with Transformers (O'Reilly)

**XIII - List of Faculties Teaching the Course**

| FACULTY NAME    | SIGNATURE |
|-----------------|-----------|
| Dr. Abdus Salam |           |

**XVIII – Verification**

|                                                                                                                                                                     |                                                                                                                                                                                    |                                                                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Prepared by:</b></p> <p>-----</p> <p><b>Dr. Abdus Salam</b><br/><i>Course Convener</i></p> <p>Date: .....</p>                                                 | <p><b>Moderated by:</b></p> <p>-----</p> <p><b>Dr. M. Mahmudul Hasan</b><br/><i>Point of Contact</i><br/><i>OBE Implementation Committee</i></p> <p>Date: .....</p>                | <p><b>Checked by:</b></p> <p>-----</p> <p><b>Prof. Dr. Muhammad Firoz Mridha</b><br/><i>Head (Undergraduate Program)</i><br/><i>Department of Computer Science</i></p> <p>Date: .....</p> |
| <p><b>Verified By:</b></p> <p>-----</p> <p><b>Dr. Md. Saef Ullah Miah</b><br/><i>Director</i><br/><i>Faculty of Science &amp; Technology</i></p> <p>Date: .....</p> | <p><b>Certified by:</b></p> <p>-----</p> <p><b>Prof. Dr. Khandaker Tabin Hasan</b><br/><i>Associate Dean</i><br/><i>Faculty of Science &amp; Technology</i></p> <p>Date: .....</p> | <p><b>Approved by:</b></p> <p>-----</p> <p><b>Prof. Dr. Dip Nandi</b><br/><i>Dean</i><br/><i>Faculty of Science &amp; Technology</i></p> <p>Date: .....</p>                               |